

(insects and allies) and may reflect an ancient and convergent solution to life on land.

Chelicerates and humans

Several aspects of human life are directly affected by the biology of chelicerates. The blood of the Atlantic horseshoe crab, *Limulus polyphemus*, is harvested in large quantities annually for the preparation of *Limulus* amoebocyte lysate, which is used for the detection of Gram-negative bacteria. Overharvesting of *L. polyphemus* has led to conservation concerns for North Atlantic populations.

Certain arachnid groups have direct impacts on human health, namely, as disease vectors, agricultural pests, and parasites. As examples, dust mites (multiple species and genera, but frequently associated with the genus *Dermatophagoides*) can cause or exacerbate allergies and asthma. The Varroa mite (*Varroa destructor*) adversely affects apiculture and severe infestations can eliminate honey bee colonies. Spider mites (Tetranychidae) are pests of some agricultural products and require management. Many species of ticks are vectors of disease. Examples of tick-borne illness include Lyme disease, tularemia, anaplasmosis, spotted fever, and Rickettsiosis.

In tropical and subtropical parts of the world, encounters with hazardous arachnids (principally spiders and scorpions) constitute a public health concern. Annually, it is estimated that 3250 people are fatally stung by scorpions, with 1000 of these in Mexico alone. Modern research efforts are aiming to harness the potency of arachnid venoms for biomedical advancement. Components of both spider and scorpion venoms are believed to hold potential for pain suppression. Scorpion venom has been used for years as an alternative treatment for cancers in parts of the Caribbean, and its potential applications for targeting tumors have gained significant attention in the past few years.

Biomedical applications of spider silk are similarly of great interest. Natural spider silk bears many desirable properties and has thus been proposed as a key target for

ongoing research efforts. Silk genes have been sequenced, isolated, and inserted into cell cultures and transgenic mammals to increase silk yield. Other approaches have targeted the production of synthetic silk, which holds the potential to surpass the properties of natural silk, exceed the strength of metals, and serve such needs as the production of protective fabrics and fulfilling uses in engineering and medicine. To date, a major challenge for both natural and synthetic spider silk remains its production in sufficient quantities for biomedical use.

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African bush elephants respond to a honeybee alarm pheromone blend

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We here report the responses of African bush elephants (*Loxodonta africana*) to a crude approximation of the honeybee alarm pheromone blend. We show that the elephants had an avoidance response to the semiochemical blend. The use of honeybee alarm pheromones to manage elephant movements in a non-invasive manner, using natural cues to which elephants may have an evolved response, holds potential for development of new options for an integrated system for elephant movement management and protection.

African elephants may inflict damage on farming areas, infrastructure, and desirable habitats in conservation areas, exacerbating wildlife-human conflict [1,2]. Conventional management action often involves killing the ‘problem animals’, often accompanied by intensified calls to reinstate elephant culling programs within conservation areas. Developing passive means of managing elephant behavior, movement and environmental impacts is thus a priority in many African countries, and other parts of the world. African elephants are strongly deterred by African honeybee (*Apis mellifera scutellata*) colonies, and methods have been developed to protect small farms and trees from elephants using bee hives [3,4]. While the use of hives is an effective natural option, there are logistic issues with maintaining large numbers of bee hives necessary for the protection of extensive fence-lines (e.g. large national parks) [1]. Elephants avoid the sight of beehives [1] and honeybee sounds [5]. However, in addition to sound and visual stimuli, African elephants have an acutely developed sense of smell, and much of their behavior is mediated by smell, rather than sound or visual stimuli [6,7].



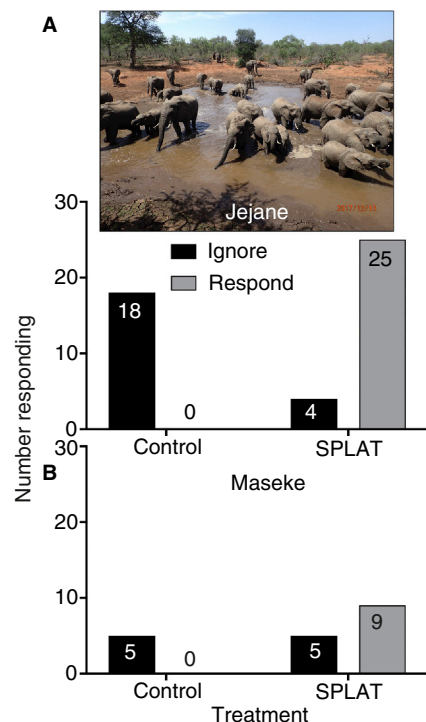


Figure 1. Elephants repelled by bee semiochemical.

Frequency of elephant responses (disturbed/repelled vs. ignored/not repelled) to honeybee alarm pheromone mix (SPLAT) and controls at Jejane (A) and Maseke (B) waterholes. Numbers of individuals in each category are shown on bars. Inset shows the Jejane study site with a visiting elephant herd.

When bee hives are disturbed, honeybees produce and release a specific blend of volatile organic compounds which serve as alarm pheromones. This communicates a threat to the colony, alerting guard bees that launch a coordinated mass attack on the potential threat. The chemical composition of the alarm pheromone complex produced by *Apis mellifera* has been analyzed [8]. Among the compounds identified in the alarm pheromone blend, some, such as iso-amyl acetate, (Z)-11-eicosen-1-ol, 2-heptanone, 2-nananol, and others have been shown to produce strong alarm responses in bees [8]. Elephants, with their highly developed sense of smell, may have either evolved or learned behavioral responses to the semiochemicals in alarm pheromones from honeybees, correlated with experience of colony attacks. We therefore investigated the response of wild *L. africana* to a prototype blend of pheromone components, formulated

into a slow-release matrix, termed SPLAT® [9]. We tested the formulation in the Associated Private Nature Reserves of the Greater Kruger National Park, Limpopo Province, South Africa.

Although the alarm pheromone of *A. mellifera scutellata* is a complex mixture of components, we decided to use a blend of two components, namely iso-amyl acetate and 2-heptanone (in equal proportions, 20% by volume of final mix), shown to produce very strong alarm responses in bees [8], formulated into a SPLAT matrix. To provide a non-threatening deployment vessel that attracted elephants' attention, we used white socks weighted with small stones; treatments comprised approximately 25 grams of SPLAT mix applied to the stones within socks, whereas controls had no semiochemical blend added. Treatments and controls were suspended from broken branches in the environment at heights of 50–100 cm above ground and placed around water holes frequented by elephants. Care was taken to ensure that wind-borne plumes from the treatments did not cross the zone downwind of the controls. Elephants visiting the waterholes were video recorded, to capture behavioral responses. We quantified the following elephant behaviors: ignoring experiment; noticing sock, approaching; showing signs of uncertainty during approach (foot swinging, trunk elevation); either moving away after signs of uncertainty or continuing to approach; touching sock; interacting with/picks up sock; ear-flap/head shake; moving away from sock. We also took note of changes in elephant behavior (e.g. stop and sniff into breeze; move away from stimulus) when they crossed semiochemical plumes on a breeze. The observations were made in December 2017 and February 2018, in Jejane Nature Reserve located 1.3 km from a site with managed beehives [4], and Maseke Game Reserve, with no known concentration of honeybees nearby. Drought conditions prevailed during these times, and elephants were regularly visiting the waterholes. We observed 55 interactions in December, and 11 in February.

We were able to discern distinct changes in behavior that were significantly associated with the presence of the pheromone blend. The elephants would either ignore controls after initial visual attraction to

the presence of an alien object in their environment, or occasionally show signs of uncertainty, pick up the controls, or try to taste them before discarding them. In the case of treatments with pheromone mix, the elephants would typically approach them, show signs of uncertainty, and then move away; in rare cases (four interactions), they would attempt to handle the treatments, but would discard them quickly. Individuals that detected the treatments in the breeze while walking would stop, turn, and extend their trunks in the direction of the stimulus, and then move away (Supplemental Video S1). On wind-still days, they appeared to need to approach the treatments to within less than one meter before responding. In the Jejane trials, 86.2% of individuals interacting with treatments showed distinct hesitation behavior or were repelled, while 0% were repelled by controls (Figure 1A; Fishers exact test, $P < 0.0001$, significant deterrence from treatments, $n = 47$ interactions). At the Maseke site, 64.3% of elephants interacting with the treatments exhibited behavior suggesting deterrence (Figure 1B, Fishers exact test, $P = 0.0217$, $n = 19$ interactions). Once at Jejane, an entire herd (27 individuals, not included in the above analyses), stopped walking and faced treatments, apparently upon detecting the breeze-born stimulus, and continued to stand and stare for a prolonged period before proceeding past the treatments.

Our results show that the elephants responded in a subtle yet distinct manner to the stimulus. They were not stimulated to bolt in fear, but showed a relatively calm response, which is a desirable degree of deterrence for such large animals. It is possible that a refined blend of semiochemicals could produce different behaviors. Elephants in the Jejane area, which had a reasonably high likelihood of encountering bee hives, showed a slightly greater (but not statistically significant, Fishers exact test $P = 0.124$) frequency of deterrence behavior than those in the Maseke area, which probably had minimal interactions with honeybees in recent years of drought conditions and low bee colony numbers in the area. We did not attempt to measure any vocalization responses [10] from the elephants in response to detection of the pheromones. Vocalization responses from the

elephants may be highly significant, as they provide a warning to the herd.

SUPPLEMENTAL INFORMATION

Supplemental Information including experimental procedures and a video can be found with this article online at <https://doi.org/10.1016/j.cub.2018.06.008>.

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Prey mistake masquerading predators for the innocuous items they resemble

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Understanding how natural selection has shaped animals' visual appearance to aid predator avoidance and prey capture has been an ongoing challenge since the conception of evolutionary theory [1,2]. Masquerade — animals resembling inedible objects common in the local environment (e.g. twigs, leaves, stones) — is one of a handful of strategies that has been suggested to serve both protective and aggressive functions (i.e. to work for both prey and predators) [3]. There is now good evidence for protective masquerade: predators detect masquerading prey but ignore them because they mistake them for the inedible objects they resemble [4]. However, there is no direct evidence that predators can benefit from aggressive masquerade [3,5]. Here, I tested the idea that prey detect masquerading predators but mistake them for the innocuous items that they resemble, making them less wary and easier for predators to catch. Because prey can only mistake masquerading predators for the objects they resemble if they have previous experience of those items, I manipulated house crickets' (*Acheta domestica*) experience with dead leaves, before placing them in tanks with dead-leaf-resembling Ghost mantises (*Phyllocrania paradoxa*). I found that mantises given crickets with experience of unmanipulated dead leaves caught crickets faster and after fewer attempts than mantises given crickets without experience of dead leaves, or crickets with experience of manipulated dead leaves that no longer resembled mantises. These findings demonstrate that predators can indeed benefit from aggressive masquerade.

The ghost mantis is a small species of mantis (4.5–5 cm) that is

widespread across Africa. Individuals of this species appear to resemble dead leaves; and given the observed intraspecific variation in color (they can be found in various shades of green and brown), and to some extent shape, it is likely that they benefit from masquerading as dead leaves (or pieces of dead leaves) from a wide variety of plant species. Their ecology is poorly characterised, but they are thought to inhabit bushes, shrubs and trees, and eat a wide variety of insect species. As ghost mantises appear to masquerade as dead leaves, I manipulated the crickets' experience with dead leaves from plants found in the same geographic area as the mantises *Coffea liberica* and *Piliostigma thonningii*. I divided the leaf-naïve crickets into three equally-sized experimental groups and gave individuals from each group a series of four experience manipulation trials. During these trials, individual crickets were placed in an arena where they were allowed to interact with either unmanipulated dead leaves, manipulated dead leaves (covered in circular white stickers) that no longer resembled mantises, or an empty arena depending on their experimental group (Supplemental Information). Crickets were initially wary of dead leaves but their latency to touch the leaves declined across trials (Supplemental Information), indicating they learned that the leaves did not pose a risk. Furthermore, there was no difference in the time to approach manipulated and unmanipulated leaves (Supplemental Information), indicating that manipulation did not influence the risk crickets perceived leaves to pose.

I then presented each of the crickets to one of thirty-six mantises, and monitored the time it took for the mantis to catch the cricket and the number of times the mantis struck at the cricket but failed to catch it (Supplemental Information). While all crickets were eventually caught, these measures differed among the three experimental groups (Latency to catch, Welch's ANOVA $F_{2,18.06} = 26.569$, $P < 0.001$. Failed attempts, Kruskal Wallis test; $\chi^2 = 22.172$, $P < 0.001$, $df = 2$; Figure 1). I found no significant differences in the time (Welch's ANOVA $F_{1,21.298} = 0.541$, $P = 0.470$; Figure 1A),

